

Chemotherapy for acute lymphoblastic leukaemia (ALL)

Chemotherapy is the main treatment for ALL. You will usually have chemotherapy as part of a clinical trial.

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What is chemotherapy?

Chemotherapy uses anti-cancer (cytotoxic) drugs to destroy or damage leukaemia cells. These drugs affect with the way [leukaemia cells](#) grow and divide.

About chemotherapy for acute lymphoblastic leukaemia (ALL)

Chemotherapy is the main treatment for [acute lymphoblastic leukaemia](#) (also called ALL or ALL leukemia). It is usually given in 3 main phases and may take up to 3 years. You have most of the drugs through a [line into a vein \(intravenously\)](#). You take others by mouth as tablets.

You will also have chemotherapy into the fluid around your spine and brain. This is called [intrathecal chemotherapy](#).

Phases of chemotherapy for ALL

The 3 main phases are:

Induction treatment



The aim of induction treatment is to quickly get rid of the leukaemia cells in your blood and bone marrow. This will allow your bone marrow to work normally again. This is called remission.

Consolidation treatment

This phase of treatment is to get rid of any remaining leukaemia cells, particularly in areas such as the brain or spinal cord.

Maintenance treatment

The aim of maintenance treatment is to prevent the leukaemia coming back and to keep you in remission.

Depending on the type of ALL you have, you may have a [targeted therapy drug or an immunotherapy drug](#) along with chemotherapy.

Some people go on to have an [allogeneic \(donor\) bone marrow transplant](#) as their best chance of a cure. This is more common if you have [Ph+ ALL](#).

If you have other health problems, you may have lower doses of chemotherapy and other drugs to control ALL for as long as possible. This is called less intensive treatment. It has less risk of [serious side effects](#) and may be easier to cope with than high doses of treatment. You may be able to have some treatments as an outpatient.

Induction treatment

This is also called induction phase or induction therapy. You usually need to stay in hospital to have your first chemotherapy treatment. You may also need to stay for a few weeks after until your blood cells recover. You will need a lot of support from [nursing and medical staff](#). They will monitor you closely for side effects, such as infection.

You may need blood or platelet transfusions because your blood cell count will be low for a few weeks. Your blood cell count will tell doctors how many red blood cells, platelets and white blood cells you have. You may also need antibiotic and antiviral drugs to prevent or treat infection. This is called [supportive therapy](#).

You may need 1 to 2 cycles of chemotherapy. You usually have a combination of 2 or 3 different drugs. Induction chemotherapy may include the following drugs:

- a [steroid](#) (prednisolone or dexamethasone), which everyone will have
- [cyclophosphamide](#)
- [vincristine](#)
- [doxorubicin](#), [daunorubicin](#) or [idarubicin](#)
- [asparaginase](#)
- [methotrexate](#)

- [rituximab](#), a targeted therapy drug, if you have [B cell ALL](#).

Testing for remission

After chemotherapy, your bone marrow and blood cells start to recover. You will have a [bone marrow biopsy](#) and [blood tests](#) to check if you are in remission or not. If tests show very small numbers of leukaemia cells (minimal residual disease) or none, your doctor will say you are in remission.

Consolidation treatment

This phase of your treatment is to get rid of any remaining leukaemia cells, particularly in areas such as the brain or spinal cord. You will have more intensive chemotherapy in hospital. But you can also have some treatments as a day patient. As with induction, you will need a lot of supportive care.

The drugs used generally include the same as those used during induction. Some other drugs that may also be used include:

- [cytarabine](#)
- [mercaptopurine](#)
- [vincristine](#).

Some people may go on to have a stem cell transplant from a donor after 2 or more cycles of chemotherapy. This depends on:

- the risk of ALL coming back
- how you are likely to cope with this treatment.

You are more likely to need a donor stem cell transplant if tests show certain genetic changes inside the leukaemia cells, such as [Ph+ ALL](#).

Chemotherapy into the spine

You will also have chemotherapy into the fluid around your spine and brain (cerebrospinal fluid). This is called [intrathecal chemotherapy](#). You have it to prevent leukaemia cells from spreading to the spine or brain (central nervous system) or to treat any leukaemia cells that may be there.

It is done using a [lumbar puncture](#). A doctor or nurse injects a small amount of chemotherapy through a thin needle into your back. You will have this done at different times during treatment for ALL.

The most common chemotherapy drugs used are:

- [methotrexate](#)
- [cytarabine](#).

You will also have a steroid drug into the spine along with the chemotherapy drugs.

Maintenance treatment

Unless you have a [donor stem cell transplant](#), you will have maintenance treatment after consolidation treatment. The aim is to prevent the leukaemia coming back and to keep you in remission. You have it for up to 2 to 3 years as an outpatient.

The most common chemotherapy combination for maintenance treatment for ALL is methotrexate and mercaptopurine. You take these as tablets.

You may also have [vincristine](#) into a vein and a [steroid](#) called prednisolone.

Side effects of chemotherapy for acute lymphoblastic leukaemia (ALL)

Chemotherapy can cause side effects. Your doctor, nurse or pharmacist will explain what to expect. This depends on the intensity of your treatment and the drugs you have. Different drugs cause different side effects. They will explain how side effects can be controlled or managed. They will also talk to you about the risk of possible [late effects](#).

Your doctors and nurses will monitor you carefully. It is important to tell them about any side effects you have.

We have more information about the [side effects of chemotherapy](#).

About our information

References



Below is a sample of the sources used in our acute lymphoblastic leukaemia (ALL) information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

NICE (National Institute for Health and Care Excellence). Blood and bone marrow cancers. Available from <https://pathways.nice.org.uk/pathways/blood-and-bone-marrow-cancers> [accessed August 2021].

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Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Senior Medical Editor, [Dr Anne Parker](#), Consultant Haematologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

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⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help